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Activity 1.4

Get a button mushroom. Draw diagram of the fruit body. Take a thin longitudinal section passing through the gill and observe the section under a microscope. Record your observations.



Dermatophytes are fungi which cause infection in skin. Example: *Trichophyton*, *Tinea*, *Microsporum* and *Epidermophyton*

The late blight disease of Potato by *Phytophthora infestans* caused a million deaths, and drove more to emigrate from Ireland (1843-1845). In India *Helminthosporium oryzae*, Blight of Paddy is also a factor for Bengal famine in 1942-1943

Activity 1.5

Keep a slice of bread in a clean plastic tray or plate. Wet the surface with little water. Leave the setup for 3 or 4 days. Observe the mouldy growth on the surface of the bread. Using a needle remove some mycelium and place it on a slide and stain the mycelium using lactophenol cotton blue. Observe the mycelium and sporangium under the microscope and Record your observation and identify the fungi and its group based on characteristic features.

1.5.7 *Rhizopus*

Class - Zygomycetes
Order - Mucorales
Family - Mucoraceae
Genus - *Rhizopus*

Rhizopus is a saprophytic fungus and grows on substrates like bread, jelly, leather, decaying vegetables and fruits. It is commonly called 'Bread mold'. *Rhizopus stolonifer* causes leak and soft rot of vegetables

Vegetative structure

The mycelium consists of aseptate, multinucleate (coenocyte) and profusely branched hyphae. There are horizontally growing aerial hyphae called **stolons**. The stolons produce rhizoids which are branched and penetrate the substratum and help in absorbing water and nutrients. Sporangiophores are borne exactly opposite to the rhizoids. The cell wall is made up of chitin and chitosan. The cell wall is followed by plasma membrane. The protoplast is granular containing many nuclei. Cell organelles like mitochondria, ribosomes and endoplasmic reticulum are present. The cell inclusions like glycogen and oil droplets are also found.

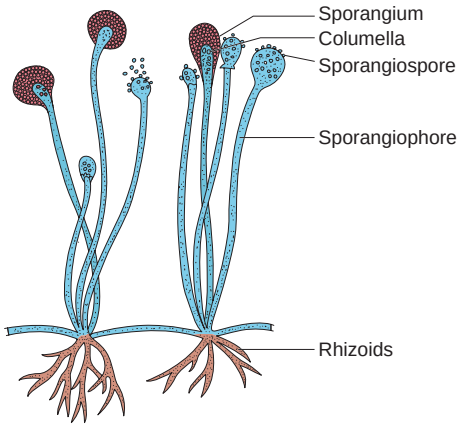


Figure 1.26: *Rhizopus*

Reproduction

Rhizopus reproduces by asexual and sexual methods.

Asexual reproduction

During favorable conditions, erect sporangiophores are produced exactly opposite to the region of formation of rhizoids of the mycelium. The sporangiophores are unicellular, unbranched